ColdTrust: LED and alert logic

What the board and the backend do in each situation. Updated 19 Oct 2026 (handling rule with a 10-minute allowance; 5 °C step; red LED is the board's own alarm).

Terms

Term	Meaning	Detail
Authorised stop	Inside an authorised geofence	Speed also under 8 km/h
Route active	"Activate Route" pressed, or any speed above 0 set while the route was idle	Status "En route". Speed 0 does not end it
Bright	Hold light at or above 30 %	Counted as the door being open
Person	The camera saw a person	Within the last 2 minutes
Out of limits	Temperature outside the limits of any package	2 readings in a row; the hold range 2-8 °C is the tightest

1. LEDs (on the board)

Situation	What happened	Result
Sensor working	A temperature reading was taken (every 5 s)	Green: short blink
Sensor failed	The DHT11 gave no reading	No green blink
Data sent	The backend received and accepted the data (every 60 s in demo, every 15 min real, or at once on an event)	Green: double blink
Data not sent	Server unreachable, or it rejected the data	No double blink
Wi-Fi down	The board lost Wi-Fi	Red: fast blink. Readings are kept for up to 1 hour and sent when Wi-Fi returns
Board alarm: bright	The hold is bright for 2 checks in a row	Red: solid
Board alarm: temperature	Temperature outside 2-8 °C for 2 readings in a row	Red: solid
All clear	The condition went away	Red: off

- Solid red wins over the fast blink.
- The LED does not know the location, so it is also lit at a dock. It is not lit by camera-only events.
- The siren, spoken message and Telegram come from the backend and dashboard, not the LED. Telegram repeats every 10 min while the problem is unresolved.

2. Alert rules (backend)

Temperature

Situation	What happened	Result
Too warm	Out of limits, warming	THERMAL_EXCURSION
Too cold	Below the limits	FREEZE_RISK
Sudden change	5 °C or more within 60 s	Critical alert. Up gives THERMAL_EXCURSION, down gives FREEZE_RISK
Steady drift	Slope of 0.5 °C per minute or more over 4 min	THERMAL_EXCURSION warning

Door and light (the hold is bright)

Situation	What happened	Result
Door ajar	Bright outside an authorised stop, moving or idle. Temperature not needed	DOOR_AJAR
Stopped on the route	Bright outside a stop, route active, speed under 2 km/h	THEFT_SUSPECTED
Handling, temperature fine	Bright at an authorised stop, temperature fine	AUTHORISED_HANDLING (logged only, no alert)
Handling, warming, short	Bright at a stop, hold warming out of limits for under 10 min	AUTHORISED_HANDLING, no message. Packages still marked under review; the budget still counts
Handling, warming, long	Same, for 10 min or more	THERMAL_EXCURSION, with Telegram, siren and voice
Handling, cooling	Bright at a stop, hold cooling (fell 5 °C or more, or below a package's minimum)	FREEZE_RISK at once. An open door never cools the hold

Person (camera)

Situation	What happened	Result
Suspicious person	Person outside an authorised stop	SUSPICIOUS_PERSON (critical, cargo under review)
Theft with a person	Person plus bright, outside an authorised stop	THEFT_SUSPECTED (emergency, packages quarantined)
Person at a stop	Person at an authorised stop	Logged as handling, no alert

Data trust

Situation	What happened	Result
Spoof	Door open for minutes but the temperature stays flat	SPOOF_SUSPECTED (admin only)
Unreliable data	Flat light reading, or the two thermometers disagree	DATA_TRUST (admin only)
Rejected message	Bad signature or a replayed message	Rejected, admin warned

3. Order of checks

The first match wins.

Check	Covers	Result
1. Theft	A person with light outside a stop, or light while stopped on the active route	THEFT_SUSPECTED
2. Spoof	Door open with no temperature response	SPOOF_SUSPECTED
3. Handling	Bright at an authorised stop (includes the 10-minute allowance for a warming hold)	AUTHORISED_HANDLING
4. Door ajar	Bright outside an authorised stop	DOOR_AJAR
5. Thermal	Out of limits, or a sudden change	THERMAL_EXCURSION or FREEZE_RISK
6. Data trust	Flat light or disagreeing thermometers	DATA_TRUST